

EMILIE BURNHAM

PhD Candidate in Astronomy & Astrophysics
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RESEARCH INTERESTS

Observational astronomer focused on exoplanet atmospheres, stellar contamination, and transmission spectroscopy with JWST.

EDUCATION

Pennsylvania State University <i>PhD in Astronomy and Astrophysics</i>	2022– State College, PA
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University of California, Santa Cruz <i>BS in Physics</i>	2018–2022 Santa Cruz, CA
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RESEARCH EXPERIENCE

Penn State University <i>Graduate Research Assistant</i> Transmission spectroscopy of exoplanet atmospheres with JWST NIRSpec and NIRISS.	2022– State College, PA
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FIRST-AUTHOR PUBLICATIONS

[1] Burnham, Emilie F., Espinoza, Nestor, Fortney, Jonathan J.. *Transmission Spectroscopy of a Sub-Neptune with JWST NIRSpec* The Astrophysical Journal Letters (2024).

CO-AUTHOR PUBLICATIONS

[1] Espinoza, Nestor, Burnham, Emilie F., Wakeford, Hannah R.. *Stellar Contamination in Transmission Spectra* The Astronomical Journal (2023).

HONORS AND AWARDS

NSF Graduate Research Fellowship <i>National Science Foundation</i>	2023
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Bunton-Waller Fellowship <i>Pennsylvania State University</i>	2022
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LEADERSHIP AND SERVICE

Graduate Student Representative <i>Penn State Department of Astronomy</i>	2023–2024 State College, PA
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OUTREACH

Astronomy on Tap Speaker <i>Astronomy on Tap State College</i>	2023 State College, PA
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PROFESSIONAL PRESENTATIONS

Invited Talks

- “*Probing Exoplanet Atmospheres with JWST Transmission Spectroscopy*”
 - Penn State University Astrophysics Seminar | State College, PA | 2024
 - University of Arizona Steward Observatory Colloquium | Tucson, AZ | 2024

Contributed Talks

- “*Stellar Contamination Corrections for JWST Transit Observations*”
 - American Astronomical Society 243rd Meeting | New Orleans, LA | 2024 | (Dissertation Talk)

Contributed Posters

- “*NIRSpec G395H Transmission Spectrum of a Sub-Neptune*”
 - JWST Early Science Symposium | Baltimore, MD | 2023

SKILLS

Programming: Python, Ruby, IDL, Bash

Software: Jupyter, LaTeX, Git, IRAF

Observing: JWST NIRSpec, NIRISS, Spitzer IRAC, ground-based spectroscopy